Puzzle

Twin primes are two prime numbers whose difference is two.

For example, 17 and 19 are twin primes.

Puzzle: Prove that for every twin prime with one prime greater than 6, the number in between the two twin primes is divisible by 6.

For example, the number between 17 and 19 is 18 which is divisible by 6.

CSEP 590tv: Quantum Computing

Dave Bacon

July 6, 2005





Today's Menu

Administrivia

Basis

Two Qubits

Deutsch's Algorithm

Begin Quantum Teleportation?

Administrivia

Hand in Homework #1

Pick up Homework #2

Is anyone not on the mailing list?

Recap

The description of a quantum system is a complex vector

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \qquad \langle v| = \begin{bmatrix} v_0^* & v_1^* & \cdots v_{N-1}^* \end{bmatrix}$$

Measurement in computational basis gives outcome with probability equal to modulus of component squared.

$$\begin{aligned} Pr(0) &= |v_0|^2 \\ Pr(1) &= |v_1|^2 \\ &\vdots \\ Pr(N) &= |v_N|^2 \end{aligned}$$

Evolution between measurements is described by a unitary matrix.

$$|v'\rangle = U|v\rangle \qquad UU^\dagger = U^\dagger U = I$$

Reca
p

Qubits:

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \end{bmatrix} = v_0|0\rangle + v_1|1\rangle$$

Measuring a qubit:

$$Pr(0) = |v_0|^2$$
$$Pr(1) = |v_1|^2$$

Unitary evolution of a qubit:

$$U = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix}$$

$$|v'\rangle = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix} \begin{bmatrix} v_0 \\ v_1 \end{bmatrix} = \begin{bmatrix} U_{0,0}v_0 + U_{0,1}v_1 \\ U_{1,0}v_0 + U_{1,1}v_1 \end{bmatrix}$$

Goal of This Lecture

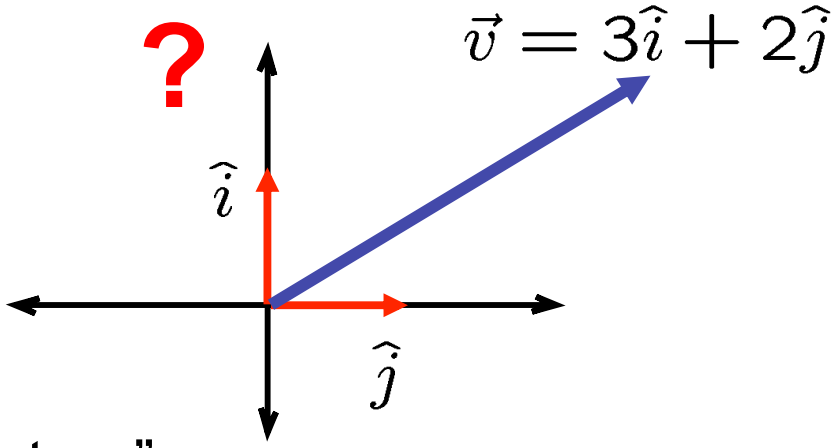
Finish off single qubits. Discuss change of basis.

Two qubits. Tensor products.

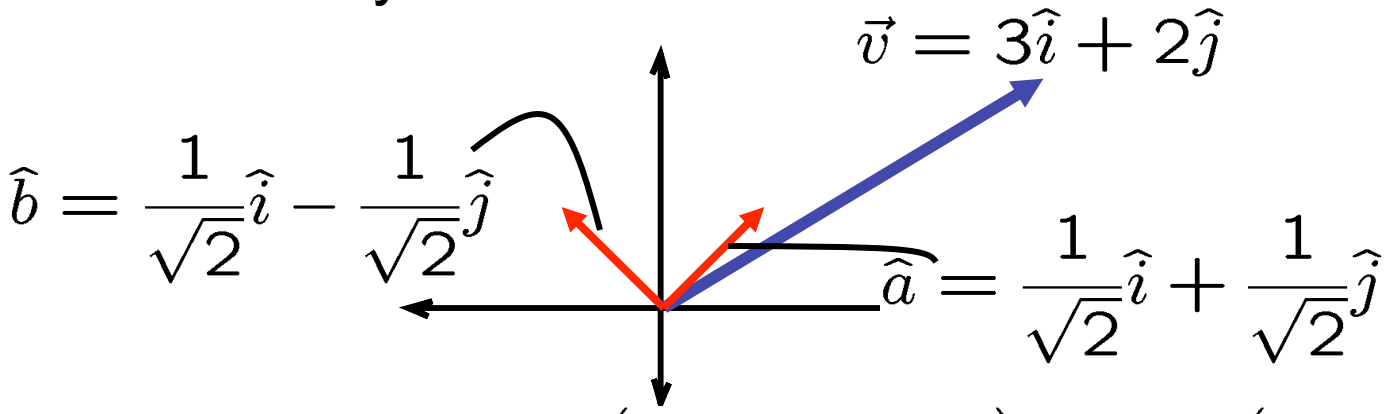
Deutsch's Problem

By the end of this lecture you will be ready to embark
on studying quantum protocols….like quantum teleportation

Basis ?



"Other coordinate system"



$$\vec{v} = \frac{5}{\sqrt{2}}\hat{a} + \frac{1}{\sqrt{2}}\hat{b} = \frac{5}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}\right) + \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}\right)$$

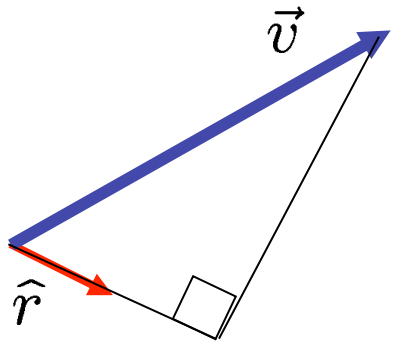
$$= \left(\frac{5}{2} + \frac{1}{2}\right)\hat{i} + \left(\frac{5}{2} - \frac{1}{2}\right)\hat{j}$$

Resolving a Vector

use the dot product to get the component of a vector
along a direction:

unit vector

$$|\hat{r}| = 1$$



$$\hat{r} \cdot \vec{v}$$

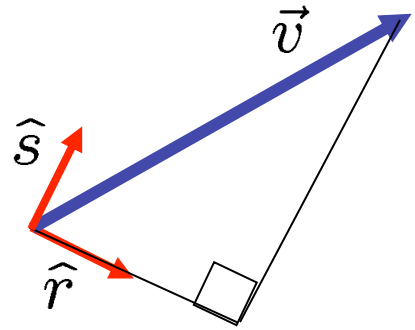
use two orthogonal unit vectors in 2D to write in new basis:

orthogonal unit vectors:

$$|\hat{r}| = 1$$

$$|\hat{s}| = 1$$

$$\hat{r} \cdot \hat{s} = 0$$

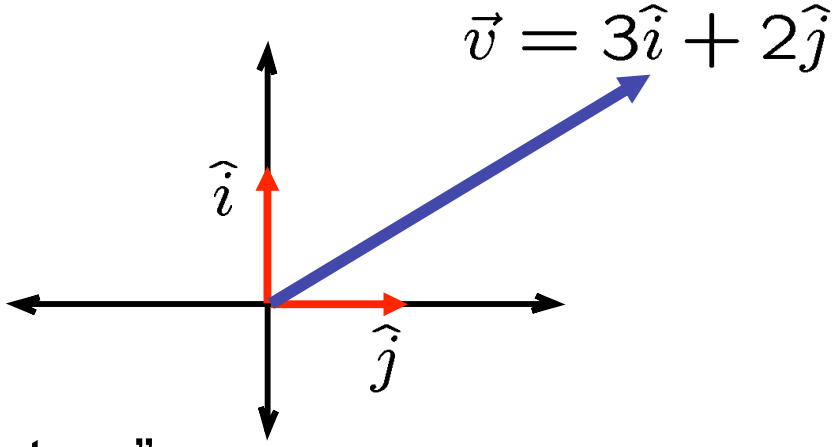


$$\hat{s} \cdot \vec{v}$$

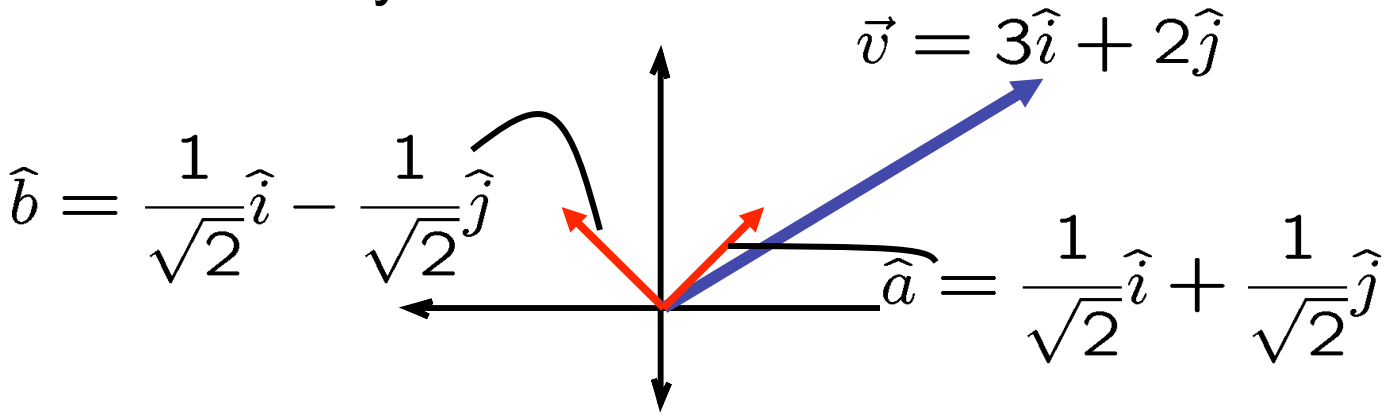
$$\vec{v} = (\hat{r} \cdot \vec{v})\hat{r} + (\hat{s} \cdot \vec{v})\hat{s}$$

$$\hat{r} \cdot \vec{v}$$

Expressing In a New Basis



"Other coordinate system"



$$\hat{a} \cdot \vec{v} = 3(\frac{1}{\sqrt{2}}) + 2(\frac{1}{\sqrt{2}}) = \frac{5}{\sqrt{2}}$$

$$\hat{b} \cdot \vec{v} = 3(\frac{1}{\sqrt{2}}) - 2(\frac{1}{\sqrt{2}}) = \frac{1}{\sqrt{2}}$$

$$\vec{v} = \frac{5}{\sqrt{2}}\hat{a} + \frac{1}{\sqrt{2}}\hat{b}$$

Computational Basis

Computational basis:

$$|0\rangle$$
$$|1\rangle$$
$$\vdots$$
$$|N-1\rangle$$

is an orthonormal basis:

$$\langle j|k\rangle = \delta_{j,k} = \begin{cases} 1 \text{ if } j = k \\ 0 \text{ otherwise} \end{cases}$$

Kronecker delta

Computational basis is important because when we measure our quantum computer (a qubit, two qubits, etc.) we get an outcome corresponding to these basis vectors.

But there are all sorts of other basis which we could use to, say, expand our vector about.

A Different Qubit Basis

A different orthonormal basis:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

$$\langle +|+\rangle = \left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{1}{\sqrt{2}}\right|^2 = 1$$

$$\langle -|-\rangle = \left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{-1}{\sqrt{2}}\right|^2 = 1$$

$$\langle +|-\rangle = \left(\frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}}\frac{-1}{\sqrt{2}}\right) = 0$$

An orthonormal basis is complete if the number of basis elements is equal to the dimension of the complex vector space.

Changing Your Basis

Express the qubit wave function $|v\rangle = v_0|0\rangle + v_1|1\rangle$ in the orthonormal complete basis

$$|a\rangle, |b\rangle$$

in other words find component of.

$$|v\rangle = v_a|a\rangle + v_b|b\rangle$$

Some inner products:

$$\langle a|v\rangle = \langle a|\left(v_a|a\rangle + v_b|b\rangle\right) = v_a\langle a|a\rangle + v_b\langle a|b\rangle = v_a$$

$$\langle b|v\rangle = \langle b|\left(v_a|a\rangle + v_b|b\rangle\right) = v_a\langle b|a\rangle + v_b\langle b|b\rangle = v_b$$

So:

$$|v\rangle = (\langle a|v\rangle)|a\rangle + (\langle b|v\rangle)|b\rangle$$

Calculating these inner products allows us to express the ket in a new basis.

Example Basis Change

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \qquad |-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

Express $|v\rangle = v_0|0\rangle + v_1|1\rangle$ in this basis: $|v\rangle = v_+|+\rangle + v_-|-\rangle$

$$\langle +|v\rangle = \langle +|\left(v_+|+\rangle + v_-|-\rangle\right) = v_+\langle +|+\rangle + v_-\langle +|-\rangle = v_+$$

$$\langle -|v\rangle = \langle -|\left(v_+|+\rangle + v_-|-\rangle\right) = v_+\langle -|+\rangle + v_-\langle -|-\rangle = v_-$$

So $|v\rangle = (\langle +|v\rangle)|+\rangle + (\langle -|v\rangle)|-\rangle$:

$$\langle +|v\rangle = \left(\frac{1}{\sqrt{2}}v_0\right) + \left(\frac{1}{\sqrt{2}}v_1\right) = \frac{v_0 + v_1}{\sqrt{2}}$$

$$\langle -|v\rangle = \left(\frac{1}{\sqrt{2}}v_0\right) + \left(\frac{-1}{\sqrt{2}}v_1\right) = \frac{v_0 - v_1}{\sqrt{2}}$$

$$|v\rangle = \frac{v_0 + v_1}{\sqrt{2}}|+\rangle + \frac{v_0 - v_1}{\sqrt{2}}|-\rangle$$

Explicit Basis Change

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \qquad |-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

Express $|v\rangle = |0\rangle$ in this basis: $|v\rangle = v_+|+\rangle + v_-|-\rangle$

$$\langle +|0\rangle = \frac{1}{\sqrt{2}}$$

$$\langle -|0\rangle = \frac{1}{\sqrt{2}}$$

$$|v\rangle = (\langle +|v\rangle)|+\rangle + (\langle -|v\rangle)|-\rangle$$

So:

$$|v\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} + \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} + \frac{1}{2} \\ \frac{1}{2} - \frac{1}{2} \end{bmatrix}$$

Basis

We can expand any vector in terms of an orthonormal basis:

$$|v\rangle = v_{\overline{0}}|\overline{0}\rangle + v_{\overline{1}}|\overline{1}\rangle + \cdots + v_{\overline{N-1}}|\overline{N-1}\rangle$$

$$\langle\overline{i}|\overline{j}\rangle = \delta_{\overline{i},\overline{j}}$$

Why does this matter? Because, as we shall see next, unitary matrices can be thought of as either rotating a vector or as a “change of basis.”

To understand this, we first note that unitary matrices have orthonormal basis already hiding within them…

Unitary Matrices, Row Vectors

$$UU^\dagger = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix} \begin{bmatrix} U^*_{0,0} & U^*_{1,0} \\ U^*_{0,1} & U^*_{1,1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Four equations:

$$\begin{bmatrix} U_{0,0}U^*_{0,0} + U_{0,1}U^*_{0,1} & U_{0,0}U^*_{1,0} + U_{0,1}U^*_{1,1} \\ U_{1,0}U^*_{0,0} + U_{1,1}U^*_{0,1} & U_{1,0}U^*_{1,0} + U_{1,1}U^*_{1,1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Say the row vectors, are an orthonormal basis

$$|\overline{0}\rangle = U_{0,0}|0\rangle + U_{0,1}|1\rangle$$

$$|\overline{1}\rangle = U_{1,0}|0\rangle + U_{1,1}|1\rangle$$

For example:

$$\begin{aligned} \langle\overline{0}|\overline{0}\rangle &= (U^*_{0,0}\langle 0| + U^*_{0,1}\langle 1|)(U_{0,0}|0\rangle + U_{0,1}|1\rangle) \\ &= U^*_{0,0}U_{0,0} + U^*_{0,1}U_{0,1} = 1 \end{aligned}$$

Unitary Matrices, Column Vectors

$$U^\dagger U = \begin{bmatrix} U^*_{0,0} & U^*_{1,0} \\ U^*_{0,1} & U^*_{1,1} \end{bmatrix} \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Four equations:

$$\begin{bmatrix} U^*_{0,0}U_{0,0} + U^*_{1,0}U_{1,0} & U^*_{0,0}U_{0,1} + U^*_{1,0}U_{1,1} \\ U^*_{0,1}U_{0,0} + U^*_{1,1}U_{1,0} & U^*_{0,1}U_{0,1} + U^*_{1,1}U_{1,1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Say the column vectors, are an orthonormal basis

$$|\overline{0}\rangle = U_{0,0}|0\rangle + U_{1,0}|1\rangle$$

$$|\overline{1}\rangle = U_{0,1}|0\rangle + U_{1,1}|1\rangle$$

For example:

$$\begin{aligned} \langle\overline{0}|\overline{0}\rangle &= (U^*_{0,0}\langle 0| + U^*_{1,0}\langle 1|)(U_{0,0}|0\rangle + U_{1,0}|1\rangle) \\ &= U^*_{0,0}U_{0,0} + U^*_{1,0}U_{1,0} = 1 \end{aligned}$$

Unitary Matrices, Row & Column

Example :

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix}$$

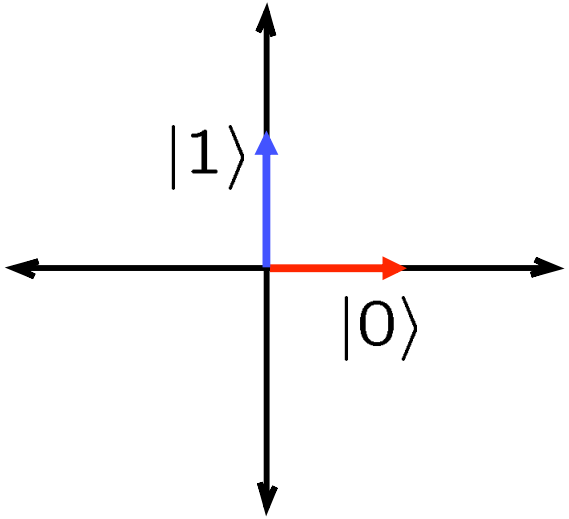
Row vectors

$$|0\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \qquad |\overline{1}\rangle = \frac{i}{\sqrt{2}}|0\rangle + \frac{-i}{\sqrt{2}}|1\rangle$$

Are orthogonal

$$\begin{aligned} \langle\overline{1}|\overline{0}\rangle &= \left(\frac{-i}{\sqrt{2}}\langle 0| + \frac{i}{\sqrt{2}}\langle 1|\right)\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \\ &= \frac{-i}{2}\langle 0|0\rangle + \frac{-i}{2}\langle 0|1\rangle + \frac{i}{2}\langle 1|0\rangle + \frac{i}{2}\langle 1|1\rangle \\ &= \frac{-i}{2} + \frac{i}{2} = 0 \end{aligned}$$

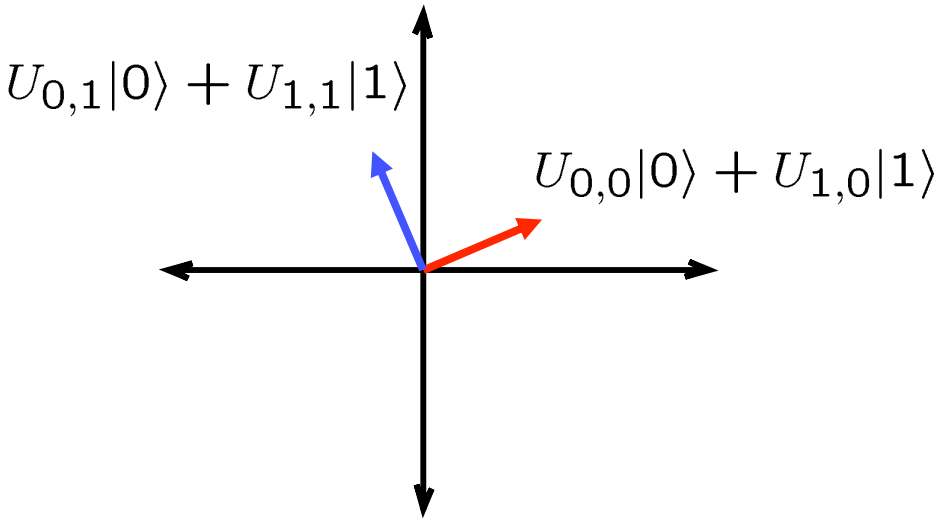
Unitary Matrices as “Rotations”



$$|0\rangle \;\boxed{U}\; U_{0,0}|0\rangle + U_{1,0}|1\rangle$$

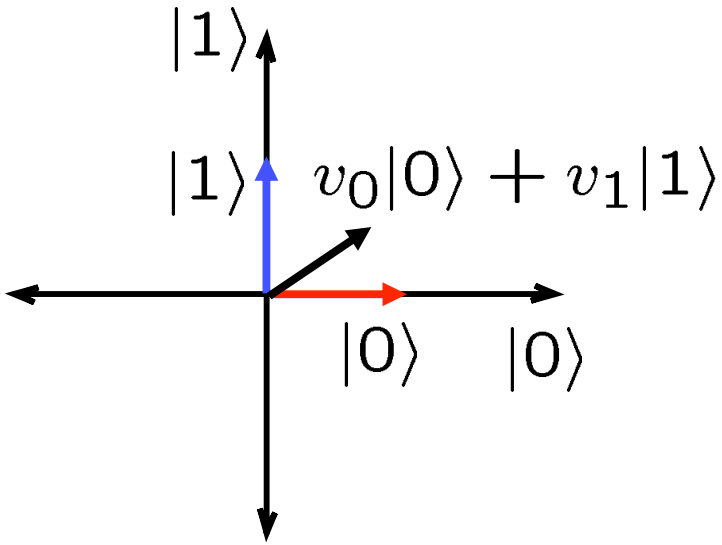
$$|1\rangle \;\boxed{U}\; U_{0,1}|0\rangle + U_{1,1}|1\rangle$$

$$U = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix}$$



Unitary matrices represent “rotations” of the complex vectors

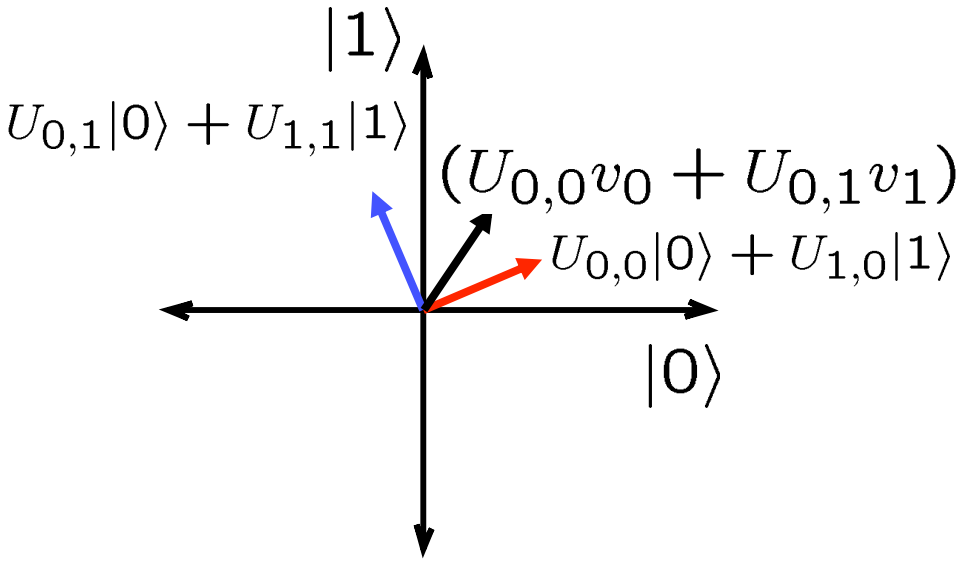
Unitary Matrices as “Rotations”



$$|0\rangle \text{ —} \boxed{U} \text{— } U_{0,0}|0\rangle + U_{1,0}|1\rangle$$

$$|1\rangle \text{ —} \boxed{U} \text{— } U_{0,1}|0\rangle + U_{1,1}|1\rangle$$

$$U = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix}$$

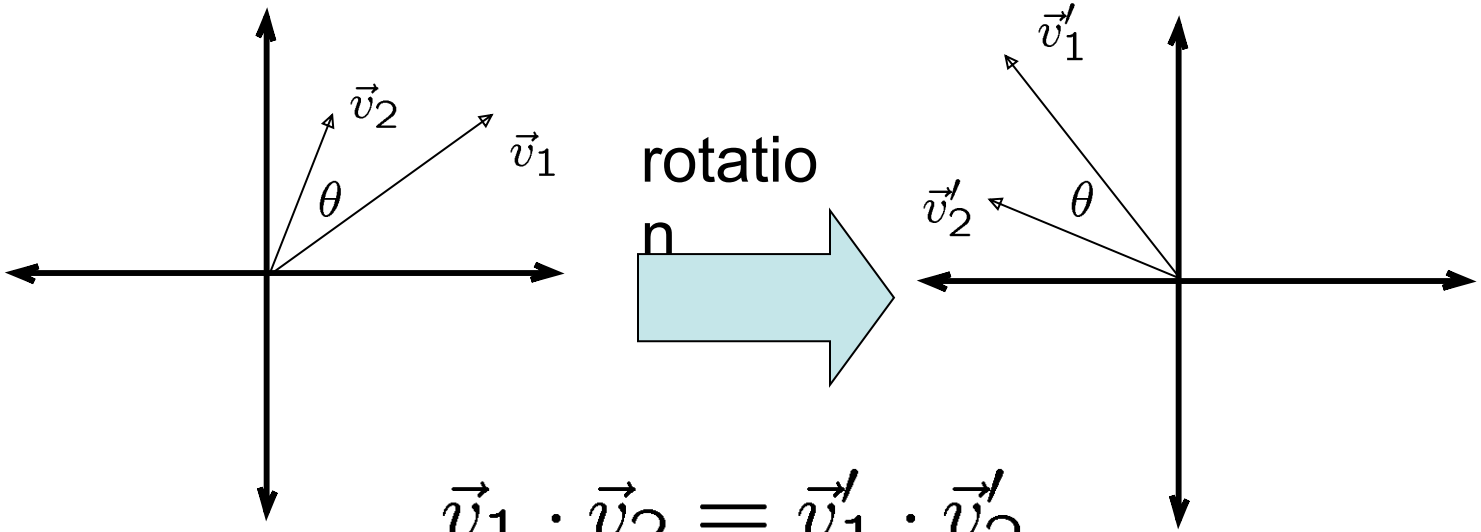


Unitary matrices represent “rotations” of the complex vectors

Rotations and Dot Products

Unitary matrices represent “rotations” of the complex vectors

Recall: rotations of real vectors preserve angles between vectors and preserve lengths of vectors.



$$\vec{v}_1 \cdot \vec{v}_2 = \vec{v}'_1 \cdot \vec{v}'_2$$
$$\vec{v}_1 \cdot \vec{v}_1 = \vec{v}'_1 \cdot \vec{v}'_1$$
$$\vec{v}_2 \cdot \vec{v}_2 = \vec{v}'_2 \cdot \vec{v}'_2$$

What is the corresponding condition for unitary matrices?

Unitary Matrices, Inner Products

Unitary matrices preserve the inner product of two complex vectors:

$$|v_1'\rangle = U|v_1\rangle$$

$$|v_2'\rangle = U|v_2\rangle$$

Adjoint-ing rule: reverse order and adjoint elements:

$$\langle v_1'| = (U|v_1\rangle)^\dagger = (|v_1\rangle)^\dagger |U^\dagger = \langle v_1|U^\dagger$$

$$\langle v_2'| = (U|v_2\rangle)^\dagger = (|v_2\rangle)^\dagger |U^\dagger = \langle v_2|U^\dagger$$

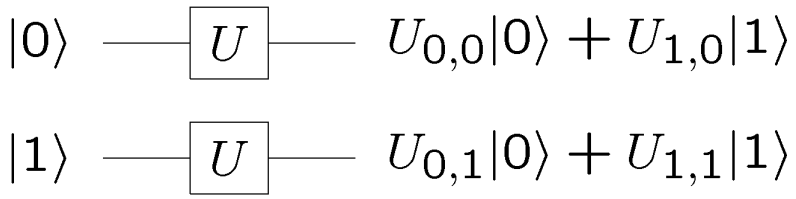
Inner product is preserved:

$$\langle v_2'|v_1'\rangle = \langle v_2|U^\dagger U|v_1\rangle = \langle v_2|I|v_1\rangle = \langle v_2|v_1\rangle$$

$$\langle v_1'|v_1'\rangle = \langle v_1|U^\dagger U|v_1\rangle = \langle v_1|I|v_1\rangle = \langle v_1|v_1\rangle$$

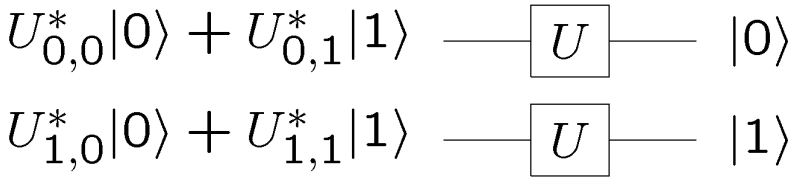
$$\langle v_2'|v_2'\rangle = \langle v_2|U^\dagger U|v_2\rangle = \langle v_2|I|v_2\rangle = \langle v_2|v_2\rangle$$

Unitary Matrices, Backwards



$$U = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix}$$

We can also ask what input vectors given computational basis vectors as their output:

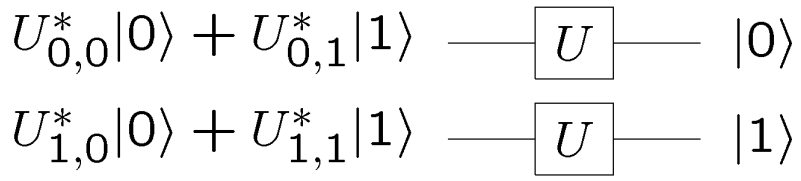


$$\begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix} \begin{bmatrix} U^*_{0,0} \\ U^*_{0,1} \end{bmatrix} = \begin{bmatrix} U_{0,0}U^*_{0,0} + U_{0,1}U^*_{0,1} \\ U_{1,0}U^*_{0,0} + U_{1,1}U^*_{0,1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

Because of unitarity:

$$UU^\dagger = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix} \begin{bmatrix} U^*_{0,0} & U^*_{1,0} \\ U^*_{0,1} & U^*_{1,1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Unitary Matrices, Basis Change



$$U = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix}$$

If we express a state $v_0|0\rangle + v_1|1\rangle$

$$U^\dagger = \begin{bmatrix} U^*_{0,0} & U^*_{1,0} \\ U^*_{0,1} & U^*_{1,1} \end{bmatrix}$$

in the row vector basis of $U^\dagger$

$$|\overline{0}\rangle = U^*_{0,0}|0\rangle + U^*_{1,0}|1\rangle \qquad |\overline{1}\rangle = U^*_{0,1}|0\rangle + U^*_{1,1}|1\rangle$$

i.e. $v_{\overline{0}}|\overline{0}\rangle + v_{\overline{1}}|\overline{1}\rangle$

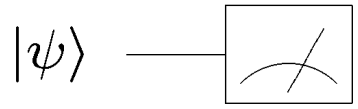
as

Then the unitary changes this state to $v_{\overline{0}}|0\rangle + v_{\overline{1}}|1\rangle$

So we can think of unitary matrices as enacting a “basis change”

Measurement Again

Recall that if we measure a qubit in the computational basis, the probability of the two outcomes 0 and 1 are



$$|\psi\rangle = v_0|0\rangle + v_1|1\rangle$$

$$Pr(0) = |v_0|^2$$

$$Pr(1) = |v_1|^2$$

We can express is in a different notation, by using

$$\langle 0|\psi\rangle = \langle 0|(v_0|0\rangle + v_1|1\rangle) = v_0$$

$$\langle 1|\psi\rangle = \langle 1|(v_0|0\rangle + v_1|1\rangle) = v_1$$

a

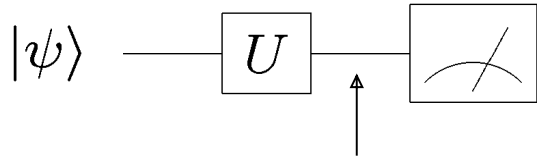
s

$$Pr(0) = |v_0|^2 = |\langle 0|\psi\rangle|^2$$

$$Pr(1) = |v_1|^2 = |\langle 1|\psi\rangle|^2$$

Unitary and Measurement

Suppose we perform a unitary evolution followed by a measurement in the computational basis:



$$|\psi'\rangle = U|\psi\rangle = v'_0|0\rangle + v'_1|1\rangle$$

What are the probabilities of the two outcomes, 0 and 1?

$$Pr(0) = |v'_0|^2 = |\langle 0|\psi'\rangle|^2 \qquad Pr(1) = |v'_1|^2 = |\langle 1|\psi'\rangle|^2$$

which we can express as

$$Pr(0) = |\langle 0|U|\psi\rangle|^2 \qquad Pr(1) = |\langle 1|U|\psi\rangle|^2$$

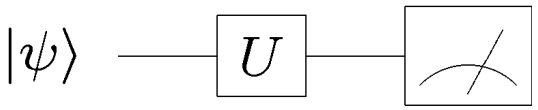
Define the new basis

$$|\bar{0}\rangle = U^\dagger|0\rangle \quad |\bar{1}\rangle = U^\dagger|1\rangle$$

Then we can express the probabilities as

$$Pr(0) = |\langle \bar{0}|\psi\rangle|^2 \qquad Pr(1) = |\langle \bar{1}|\psi\rangle|^2$$

Measurement in a Basis



$$Pr(0) = |\langle\bar{0}|\psi\rangle|^2$$
$$Pr(1) = |\langle\bar{1}|\psi\rangle|^2$$

The unitary transform allows to "perform a measurement in a basis differing from the computational basis":

$$|\bar{0}\rangle = U^\dagger|0\rangle$$
$$|\bar{1}\rangle = U^\dagger|1\rangle$$

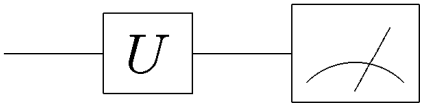
Suppose $|\bar{0}\rangle, |\bar{1}\rangle, \ldots, |\overline{N-1}\rangle$ is a complete basis. Then we can
"perform a measurement in this basis" and obtain outcomes with probabilities given by:

$$Pr(0) = |\langle\bar{0}|\psi\rangle|^2$$
$$Pr(1) = |\langle\bar{1}|\psi\rangle|^2$$
$$\vdots$$
$$Pr(N-1) = |\langle\overline{N-1}|\psi\rangle|^2$$

Measurement in a Basis

Example:

$$|\psi\rangle = v_0|0\rangle + v_1|1\rangle$$



$$Pr(0) = |\langle\bar{0}|\psi\rangle|^2$$
$$Pr(1) = |\langle\bar{1}|\psi\rangle|^2$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix} \qquad U^\dagger = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \end{bmatrix}$$

$$|\bar{0}\rangle = U^\dagger|0\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \quad |\bar{1}\rangle = U^\dagger|1\rangle = \frac{i}{\sqrt{2}}|0\rangle + \frac{-i}{\sqrt{2}}|1\rangle$$

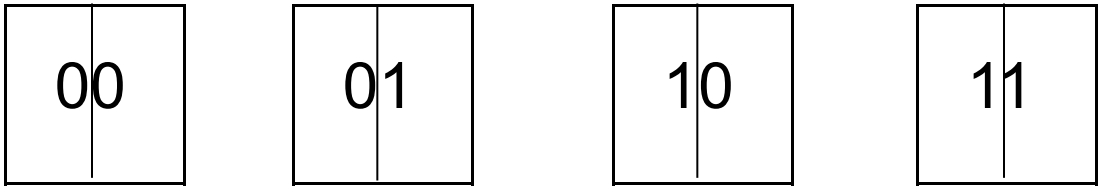
$$Pr(0) = |\langle\bar{0}|\psi\rangle|^2 = \left|(\frac{1}{\sqrt{2}}\langle 0| + \frac{1}{\sqrt{2}}\langle 1|)|\psi\rangle\right|^2 = \left|\frac{v_0}{\sqrt{2}} + \frac{v_1}{\sqrt{2}}\right|^2$$

$$Pr(1) = |\langle\bar{1}|\psi\rangle|^2 = \left|(\frac{-i}{\sqrt{2}}\langle 0| + \frac{i}{\sqrt{2}}\langle 1|)|\psi\rangle\right|^2 = \left|\frac{-iv_0}{\sqrt{2}} + \frac{iv_1}{\sqrt{2}}\right|^2$$

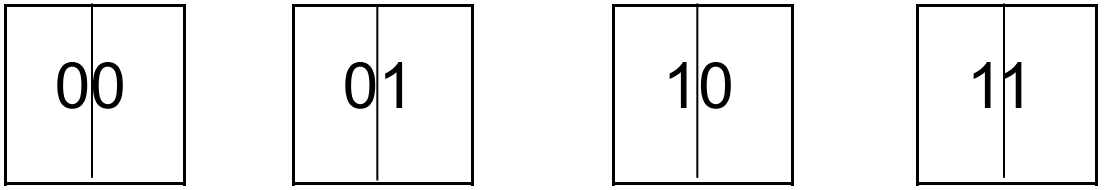
In Class Problem #1

Two Qubits

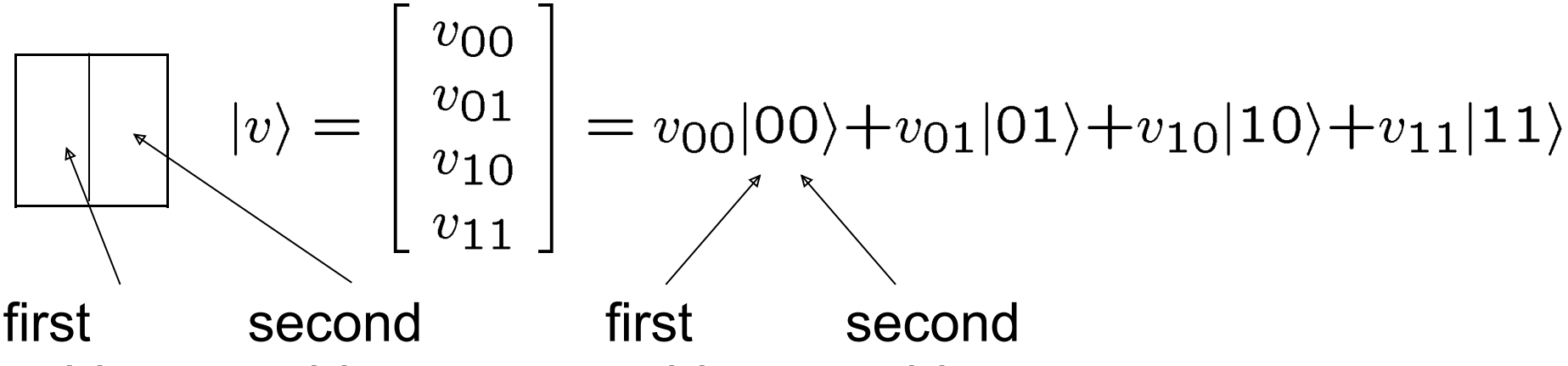
Two bits can be in one of four different states



Similarly two qubits have four different states



The wave function for two qubits thus has four components:



Two Qubits

Examples:

$$|v\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \\ 0 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}|00\rangle + \frac{i}{\sqrt{2}}|01\rangle$$

$$|v\rangle = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|11\rangle$$

$$|v\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = |00\rangle$$

When Two Qubits Are Two

The wave function for two qubits has four components:

$$|v\rangle = \begin{bmatrix} v_{00} \\ v_{01} \\ v_{10} \\ v_{11} \end{bmatrix} = v_{00}|00\rangle + v_{01}|01\rangle + v_{10}|10\rangle + v_{11}|11\rangle$$

Sometimes we can write the wave function of two qubits as the "tensor product" of two one qubit wave functions.

$|v\rangle = |a\rangle \otimes |b\rangle$ "separable"

$$|a\rangle = a_0|0\rangle + a_1|1\rangle \qquad |b\rangle = b_0|0\rangle + b_1|1\rangle$$

$$|v\rangle = (a_0|0\rangle + a_1|1\rangle) \otimes (b_0|0\rangle + b_1|1\rangle)$$

$$= a_0b_0|0\rangle\otimes|0\rangle + a_0b_1|0\rangle\otimes|1\rangle + a_1b_0|1\rangle\otimes|0\rangle + a_1b_1|1\rangle\otimes|1\rangle$$

$$= a_0b_0|00\rangle + a_0b_1|01\rangle + a_1b_0|10\rangle + a_1b_1|11\rangle$$

Two Qubits, Separable

Example : $|v\rangle = |a\rangle \otimes |b\rangle$

$$|a\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle \quad |b\rangle = \frac{i}{\sqrt{5}}|0\rangle + \frac{2}{\sqrt{5}}|1\rangle$$

$$|v\rangle = |a\rangle \otimes |b\rangle = \left(\frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle\right) \otimes \left(\frac{i}{\sqrt{5}}|0\rangle + \frac{2}{\sqrt{5}}|1\rangle\right)$$

$$= \frac{1}{2}\frac{i}{\sqrt{5}}|0\rangle \otimes |0\rangle + \frac{1}{2}\frac{2}{\sqrt{5}}|0\rangle \otimes |1\rangle + \frac{\sqrt{3}}{2}\frac{i}{\sqrt{5}}|1\rangle \otimes |0\rangle + \frac{\sqrt{3}}{2}\frac{2}{\sqrt{5}}|1\rangle \otimes |1\rangle$$

$$= \frac{i}{2\sqrt{5}}|00\rangle + \frac{1}{\sqrt{5}}|01\rangle + \frac{\sqrt{3}i}{2\sqrt{5}}|10\rangle + \frac{\sqrt{3}}{\sqrt{5}}|11\rangle$$

Two Qubits, Entangled

Example:

$$|v\rangle = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|11\rangle$$

Assume

$|v\rangle = |a\rangle \otimes |b\rangle$ $\quad |a\rangle = a_0|0\rangle + a_1|1\rangle \quad |b\rangle = b_0|0\rangle + b_1|1\rangle$

$$= a_0b_0|00\rangle + a_0b_1|01\rangle + a_1b_0|10\rangle + a_1b_1|11\rangle$$

$a_0b_1 = 0$

Either

$a_0 = 0$ but this implies $a_0b_0 = 0$

or

$b_1 = 0$ but this implies $a_1b_1 = 0$

contradictions

So $|v\rangle$ is not a separable state. It is entangled

Measuring Two Qubits

If we measure both qubits in the computational basis, then we get one of four outcomes: 00, 01, 10, and 11

If the wave function for the two qubits is

$$|v\rangle = \begin{bmatrix} v_{00} \\ v_{01} \\ v_{10} \\ v_{11} \end{bmatrix} = v_{00}|00\rangle + v_{01}|01\rangle + v_{10}|10\rangle + v_{11}|11\rangle$$

Probability of 00 is $|v_{00}|^2$ New wave function is $|00\rangle$

Probability of 01 is $|v_{01}|^2$ New wave function is $|01\rangle$

Probability of 10 is $|v_{10}|^2$ New wave function is $|10\rangle$

Probability of 11 is $|v_{11}|^2$ New wave function is $|11\rangle$

Two Qubits, Measuring

Example:

$$|v\rangle = \frac{i}{2\sqrt{5}}|00\rangle + \frac{1}{\sqrt{5}}|01\rangle + \frac{\sqrt{3}i}{2\sqrt{5}}|10\rangle + \frac{\sqrt{3}}{\sqrt{5}}|11\rangle$$

Probability of 00 is $|v_{00}|^2 = \left|\frac{i}{2\sqrt{5}}\right|^2 = \frac{1}{20}$

Probability of 01 is $|v_{01}|^2 = \left|\frac{1}{\sqrt{5}}\right|^2 = \frac{1}{5}$

Probability of 10 is $|v_{10}|^2 = \left|\frac{\sqrt{3}i}{2\sqrt{6}}\right|^2 = \frac{3}{20}$

Probability of 11 is $|v_{11}|^2 = \left|\frac{\sqrt{3}}{\sqrt{5}}\right|^2 = \frac{3}{5}$

Two Qubit Evolutions

Rule 2: The wave function of a N dimensional quantum system evolves in time according to a unitary matrix U. If the wave function initially is $|v\rangle$ then after the evolution correspond to U the new wave function is $|v'\rangle = U|v\rangle$

$$U = \begin{bmatrix} U_{00,00} & U_{00,01} & U_{00,10} & U_{00,11} \\ U_{01,00} & U_{01,01} & U_{01,10} & U_{01,11} \\ U_{10,00} & U_{10,01} & U_{10,10} & U_{10,11} \\ U_{11,00} & U_{11,01} & U_{11,10} & U_{11,11} \end{bmatrix} \qquad |v\rangle = \begin{bmatrix} v_{00} \\ v_{01} \\ v_{10} \\ v_{11} \end{bmatrix}$$

$$|v'\rangle = U|v\rangle$$

$$|v'\rangle = \begin{bmatrix} U_{00,00}v_{00} + U_{00,01}v_{01} + U_{00,10}v_{10} + U_{00,11}v_{11} \\ U_{01,00}v_{00} + U_{01,01}v_{01} + U_{01,10}v_{10} + U_{01,11}v_{11} \\ U_{10,00}v_{00} + U_{10,01}v_{01} + U_{10,10}v_{10} + U_{10,11}v_{11} \\ U_{11,00}v_{00} + U_{11,01}v_{01} + U_{11,10}v_{10} + U_{11,11}v_{11} \end{bmatrix}$$

Two Qubit Evolutions

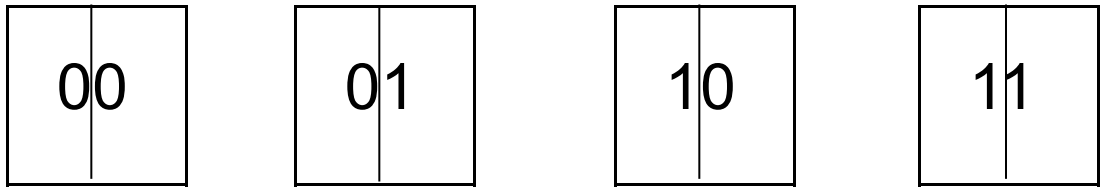
$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 & 0 \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad |v\rangle = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|11\rangle$$

$$|v'\rangle = \begin{bmatrix} U_{00,00}v_{00} + U_{00,01}v_{01} + U_{00,10}v_{10} + U_{00,11}v_{11} \\ U_{01,00}v_{00} + U_{01,01}v_{01} + U_{01,10}v_{10} + U_{01,11}v_{11} \\ U_{10,00}v_{00} + U_{10,01}v_{01} + U_{10,10}v_{10} + U_{10,11}v_{11} \\ U_{11,00}v_{00} + U_{11,01}v_{01} + U_{11,10}v_{10} + U_{11,11}v_{11} \end{bmatrix}$$

$$|v'\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}}\frac{1}{2} + \frac{i}{\sqrt{2}} \cdot 0 + 0 \cdot 0 + 0 \cdot \frac{\sqrt{3}}{2} \\ \frac{i}{\sqrt{2}}\frac{1}{2} + \frac{1}{\sqrt{2}} \cdot 0 + 0 \cdot 0 + 0 \cdot \frac{\sqrt{3}}{2} \\ 0 \cdot \frac{1}{2} + 0 \cdot 0 + 0 \cdot 0 + 1 \cdot \frac{\sqrt{3}}{2} \\ 0 \cdot \frac{1}{2} + 0 \cdot 0 + 1 \cdot 0 + 0 \cdot \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2\sqrt{2}} \\ \frac{i}{2\sqrt{2}} \\ \frac{\sqrt{3}}{2} \\ 0 \end{bmatrix}$$

Manipulations of Two Bits

Two bits can be in one of four different states



We can manipulate these bits

$$00 \longrightarrow 01$$
$$01 \longrightarrow 00$$
$$10 \longrightarrow 10$$
$$11 \longrightarrow 11$$

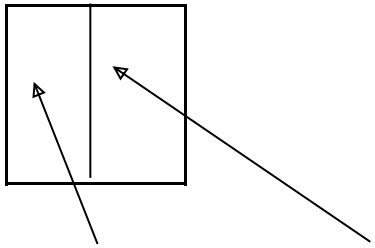
Sometimes this can be thought of as just operating on one of the bits (for example, flip the second bit):

$$00 \longrightarrow 01$$
$$01 \longrightarrow 00$$
$$10 \longrightarrow 11$$
$$11 \longrightarrow 10$$

But sometimes we cannot (as in the first example above)

Manipulations of Two Qubits

Similarly, we can apply unitary operations on only one of the qubits at a time:



first qubit

second qubit

Unitary operator that acts only on the first qubit

$$U = V \otimes I$$

two dimensional unitary matrix

two dimensional Identity matrix

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Unitary operator that acts only on the second qubit:

$$U = I \otimes V$$

Tensor Product of Matrices

$$U = V \otimes W \qquad V = \begin{bmatrix} V_{0,0} & V_{0,1} \\ V_{1,0} & V_{1,1} \end{bmatrix} \quad W = \begin{bmatrix} W_{0,0} & W_{0,1} \\ W_{1,0} & W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} V_{0,0} & V_{0,1} \\ V_{1,0} & V_{1,1} \end{bmatrix} \otimes \begin{bmatrix} W_{0,0} & W_{0,1} \\ W_{1,0} & W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} V_{0,0}W_{0,0} & V_{0,0}W_{0,1} & V_{0,1}W_{0,0} & V_{0,1}W_{0,1} \\ V_{0,0}W_{1,0} & V_{0,0}W_{1,1} & V_{0,1}W_{1,0} & V_{0,1}W_{1,1} \\ V_{1,0}W_{0,0} & V_{1,0}W_{0,1} & V_{1,1}W_{0,0} & V_{1,1}W_{0,1} \\ V_{1,0}W_{1,0} & V_{1,0}W_{1,1} & V_{1,1}W_{1,0} & V_{1,1}W_{1,1} \end{bmatrix}$$

Tensor Product of Matrices

Example: $U = V \otimes W$

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \quad W = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$$

$$U = \begin{bmatrix} V_{0,0}W_{0,0} & V_{0,0}W_{0,1} & V_{0,1}W_{0,0} & V_{0,1}W_{0,1} \\ V_{0,0}W_{1,0} & V_{0,0}W_{1,1} & V_{0,1}W_{1,0} & V_{0,1}W_{1,1} \\ V_{1,0}W_{0,0} & V_{1,0}W_{0,1} & V_{1,1}W_{0,0} & V_{1,1}W_{0,1} \\ V_{1,0}W_{1,0} & V_{1,0}W_{1,1} & V_{1,1}W_{1,0} & V_{1,1}W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}}W_{0,0} & \frac{1}{\sqrt{2}}W_{0,1} & \frac{i}{\sqrt{2}}W_{0,0} & \frac{i}{\sqrt{2}}W_{0,1} \\ \frac{1}{\sqrt{2}}W_{1,0} & \frac{1}{\sqrt{2}}W_{1,1} & \frac{i}{\sqrt{2}}W_{1,0} & \frac{i}{\sqrt{2}}W_{1,1} \\ \frac{i}{\sqrt{2}}W_{0,0} & \frac{i}{\sqrt{2}}W_{0,1} & \frac{1}{\sqrt{2}}W_{0,0} & \frac{1}{\sqrt{2}}W_{0,1} \\ \frac{i}{\sqrt{2}}W_{1,0} & \frac{i}{\sqrt{2}}W_{1,1} & \frac{1}{\sqrt{2}}W_{1,0} & \frac{1}{\sqrt{2}}W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}}\frac{1}{2} & \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} & \frac{i}{\sqrt{2}}\frac{1}{2} & \frac{i}{\sqrt{2}}\frac{\sqrt{3}}{2} \\ \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} & \frac{1}{\sqrt{2}}\frac{-1}{2} & \frac{i}{\sqrt{2}}\frac{\sqrt{3}}{2} & \frac{i}{\sqrt{2}}\frac{-1}{2} \\ \frac{i}{\sqrt{2}}\frac{1}{2} & \frac{i}{\sqrt{2}}\frac{\sqrt{3}}{2} & \frac{1}{\sqrt{2}}\frac{1}{2} & \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} \\ \frac{i}{\sqrt{2}}\frac{\sqrt{3}}{2} & \frac{i}{\sqrt{2}}\frac{-1}{2} & \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} & \frac{1}{\sqrt{2}}\frac{-1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2\sqrt{2}} & \frac{\sqrt{3}}{2\sqrt{2}} & \frac{i}{2\sqrt{2}} & \frac{\sqrt{3}i}{2\sqrt{2}} \\ \frac{\sqrt{3}}{2\sqrt{2}} & \frac{-1}{2\sqrt{2}} & \frac{\sqrt{3}i}{2\sqrt{2}} & \frac{-i}{2\sqrt{2}} \\ \frac{i}{2\sqrt{2}} & \frac{\sqrt{3}i}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & \frac{\sqrt{3}}{2\sqrt{2}} \\ \frac{\sqrt{3}i}{2\sqrt{2}} & \frac{-i}{2\sqrt{2}} & \frac{\sqrt{3}}{2\sqrt{2}} & \frac{-1}{2\sqrt{2}} \end{bmatrix}$$

Tensor Product of Matrices

Example: $U = V \otimes I$

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \qquad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} V_{0,0}W_{0,0} & V_{0,0}W_{0,1} & V_{0,1}W_{0,0} & V_{0,1}W_{0,1} \\ V_{0,0}W_{1,0} & V_{0,0}W_{1,1} & V_{0,1}W_{1,0} & V_{0,1}W_{1,1} \\ V_{1,0}W_{0,0} & V_{1,0}W_{0,1} & V_{1,1}W_{0,0} & V_{1,1}W_{0,1} \\ V_{1,0}W_{1,0} & V_{1,0}W_{1,1} & V_{1,1}W_{1,0} & V_{1,1}W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}}1 & \frac{1}{\sqrt{2}}0 & \frac{i}{\sqrt{2}}1 & \frac{i}{\sqrt{2}}0 \\ \frac{1}{\sqrt{2}}0 & \frac{1}{\sqrt{2}}1 & \frac{i}{\sqrt{2}}0 & \frac{i}{\sqrt{2}}1 \\ \frac{i}{\sqrt{2}}1 & \frac{i}{\sqrt{2}}0 & \frac{1}{\sqrt{2}}1 & \frac{1}{\sqrt{2}}0 \\ \frac{i}{\sqrt{2}}0 & \frac{i}{\sqrt{2}}1 & \frac{1}{\sqrt{2}}0 & \frac{1}{\sqrt{2}}1 \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{i}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{i}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Tensor Product of Matrices

Example: $U = I \otimes W$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad W = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U = \begin{bmatrix} V_{0,0}W_{0,0} & V_{0,0}W_{0,1} & V_{0,1}W_{0,0} & V_{0,1}W_{0,1} \\ V_{0,0}W_{1,0} & V_{0,0}W_{1,1} & V_{0,1}W_{1,0} & V_{0,1}W_{1,1} \\ V_{1,0}W_{0,0} & V_{1,0}W_{0,1} & V_{1,1}W_{0,0} & V_{1,1}W_{0,1} \\ V_{1,0}W_{1,0} & V_{1,0}W_{1,1} & V_{1,1}W_{1,0} & V_{1,1}W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} 1\frac{1}{\sqrt{2}} & 1\frac{i}{\sqrt{2}} & 0\frac{1}{\sqrt{2}} & 0\frac{i}{\sqrt{2}} \\ 1\frac{i}{\sqrt{2}} & 1\frac{1}{\sqrt{2}} & 0\frac{i}{\sqrt{2}} & 0\frac{1}{\sqrt{2}} \\ 0\frac{1}{\sqrt{2}} & 0\frac{i}{\sqrt{2}} & 1\frac{1}{\sqrt{2}} & 1\frac{i}{\sqrt{2}} \\ 0\frac{i}{\sqrt{2}} & 0\frac{1}{\sqrt{2}} & 1\frac{i}{\sqrt{2}} & 1\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 & 0 \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ 0 & 0 & \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Tensor Product of Matrices

Example: $U = I \otimes W$

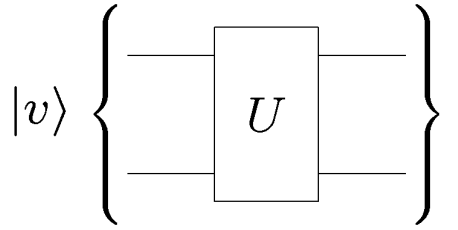
$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad W = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U = \begin{bmatrix} V_{0,0}W_{0,0} & V_{0,0}W_{0,1} & V_{0,1}W_{0,0} & V_{0,1}W_{0,1} \\ V_{0,0}W_{1,0} & V_{0,0}W_{1,1} & V_{0,1}W_{1,0} & V_{0,1}W_{1,1} \\ V_{1,0}W_{0,0} & V_{1,0}W_{0,1} & V_{1,1}W_{0,0} & V_{1,1}W_{0,1} \\ V_{1,0}W_{1,0} & V_{1,0}W_{1,1} & V_{1,1}W_{1,0} & V_{1,1}W_{1,1} \end{bmatrix}$$

$$U = \begin{bmatrix} 1\frac{1}{\sqrt{2}} & 1\frac{i}{\sqrt{2}} & 0\frac{1}{\sqrt{2}} & 0\frac{i}{\sqrt{2}} \\ 1\frac{i}{\sqrt{2}} & 1\frac{1}{\sqrt{2}} & 0\frac{i}{\sqrt{2}} & 0\frac{1}{\sqrt{2}} \\ 0\frac{1}{\sqrt{2}} & 0\frac{i}{\sqrt{2}} & 1\frac{1}{\sqrt{2}} & 1\frac{i}{\sqrt{2}} \\ 0\frac{i}{\sqrt{2}} & 0\frac{1}{\sqrt{2}} & 1\frac{i}{\sqrt{2}} & 1\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 & 0 \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ 0 & 0 & \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

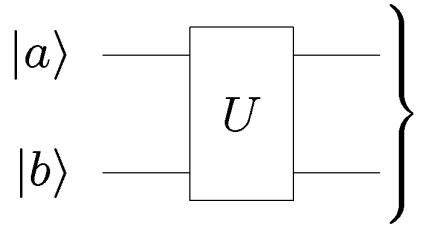
Two Qubit Quantum Circuits



$$|v'\rangle = U|v\rangle$$

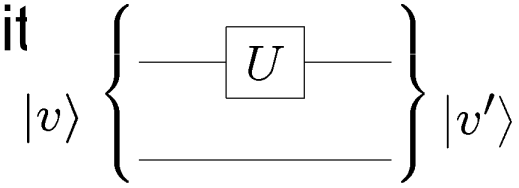
A two qubit unitary gate

Sometimes the input our output is known to be seperable:

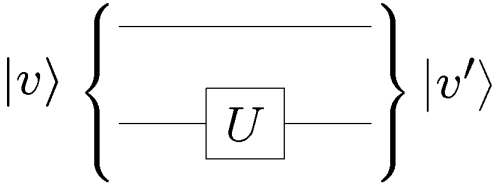


$$|v'\rangle = U(|a\rangle \otimes |b\rangle)$$

Sometimes we act only one qubit



$$|v'\rangle = (U \otimes I)|v\rangle$$

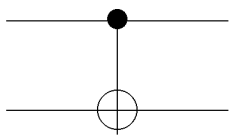


$$|v'\rangle = (I \otimes U)|v\rangle$$

Some Two Qubit Gates

control

target



controlled-NOT $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

$$|00\rangle \rightarrow |00\rangle$$

$$|01\rangle \rightarrow |01\rangle$$

$$|10\rangle \rightarrow |11\rangle$$

$$|11\rangle \rightarrow |10\rangle$$

Conditional on the first bit, the gate flips the second bit.

Computational Basis and Unitaries

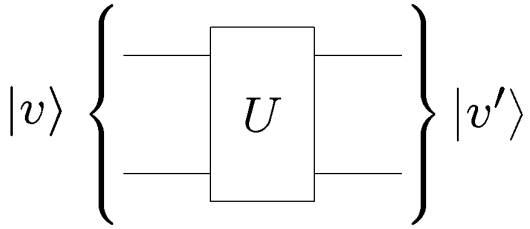
$$U = \begin{bmatrix} U_{00,00} & U_{00,01} & U_{00,10} & U_{00,11} \\ U_{01,00} & U_{01,01} & U_{01,10} & U_{01,11} \\ U_{10,00} & U_{10,01} & U_{10,10} & U_{10,11} \\ U_{11,00} & U_{11,01} & U_{11,10} & U_{11,11} \end{bmatrix} \qquad |00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$|v'\rangle = U|00\rangle = v'_{00}|00\rangle + v'_{01}|01\rangle + v'_{10}|10\rangle + v'_{11}|11\rangle$$

$$|v'\rangle = \begin{bmatrix} U_{00,00}v_{00} + U_{00,01}v_{01} + U_{00,10}v_{10} + U_{00,11}v_{11} \\ U_{01,00}v_{00} + U_{01,01}v_{01} + U_{01,10}v_{10} + U_{01,11}v_{11} \\ U_{10,00}v_{00} + U_{10,01}v_{01} + U_{10,10}v_{10} + U_{10,11}v_{11} \\ U_{11,00}v_{00} + U_{11,01}v_{01} + U_{11,10}v_{10} + U_{11,11}v_{11} \end{bmatrix}$$

Notice that by examining the unitary evolution of all computational basis states, we can explicitly determine what the unitary matrix.

Linearity



$$|v\rangle = v_{00}|00\rangle + v_{01}|01\rangle + v_{10}|10\rangle + v_{11}|11\rangle$$

$$|v'\rangle = U|v\rangle = v_{00}U|00\rangle + v_{01}U|01\rangle + v_{10}U|10\rangle + v_{11}U|11\rangle$$

We can act on each computational basis state and then resum

This simplifies calculations considerably

Linearity

Example:

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 & 0 \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad |v\rangle = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|11\rangle$$

$$|v'\rangle = U|v\rangle = \frac{1}{2}U|00\rangle + \frac{\sqrt{3}}{2}U|11\rangle$$

$$= \frac{1}{2}\left(\frac{1}{\sqrt{2}}|00\rangle + \frac{i}{\sqrt{2}}|01\rangle\right) + \frac{\sqrt{3}}{2}|10\rangle$$

$$= \frac{1}{2\sqrt{2}}|00\rangle + \frac{i}{2\sqrt{2}}|01\rangle + \frac{\sqrt{3}}{2}|10\rangle$$

Linearity

Example:

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 & 0 \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad |v\rangle = \frac{1}{2}|00\rangle + \frac{\sqrt{3}}{2}|01\rangle$$

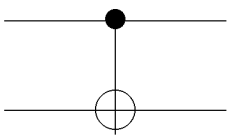
$$|v'\rangle = U|v\rangle = \frac{1}{2}U|00\rangle + \frac{\sqrt{3}}{2}U|01\rangle$$

$$= \frac{1}{2}\left(\frac{1}{\sqrt{2}}|00\rangle + \frac{i}{\sqrt{2}}|01\rangle\right) + \frac{\sqrt{3}}{2}\left(\frac{i}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|01\rangle\right)$$

$$= \frac{1 + i\sqrt{3}}{2\sqrt{2}}|00\rangle + \frac{\sqrt{3} + i}{2\sqrt{2}}|01\rangle$$

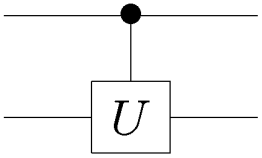
Some Two Qubit Gates

control
target

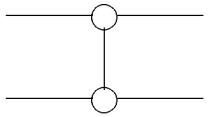


controlled-NOT $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

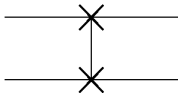
control
target



controlled-U $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{00} & U_{01} \\ 0 & 0 & U_{10} & U_{11} \end{bmatrix}$



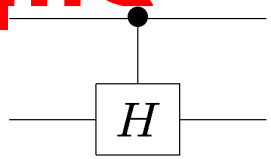
controlled-phase $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$



swap $= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

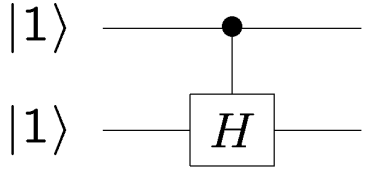
Quantum Circuits

controlled-H



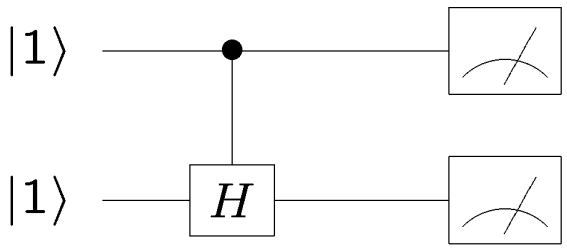
$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$|v\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = |11\rangle$$



$$|v'\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}}(|10\rangle - |11\rangle)$$



Probability of 10: $\left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

Probability of 11: $\left|\frac{-1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

Probability of 00 and 01: $|0|^2 = 0$

In Class Problem #2